

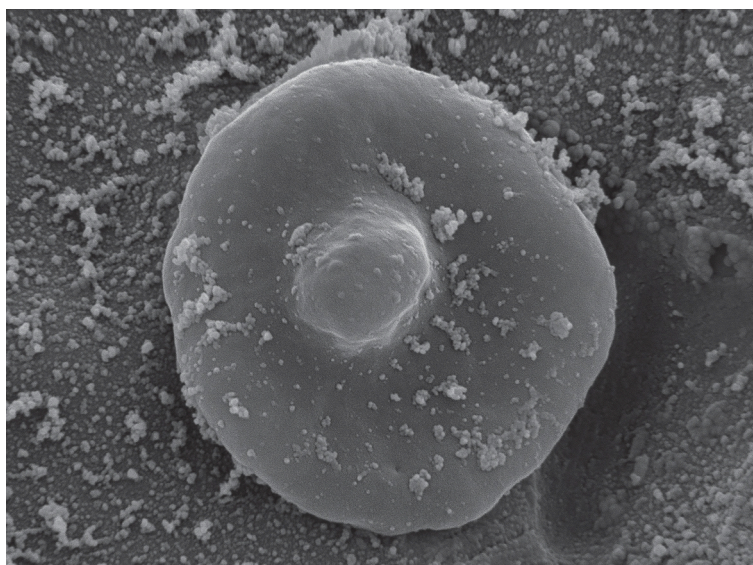
- 4 A number of species in the genus *Plasmodium* are known to cause malaria in humans.

Plasmodium has a complex life cycle with a number of different structural forms. The merozoite is one form of the pathogen that is present in human hosts.

- (a) State the type of organism that causes malaria.

..... [1]

Fig. 4.1 is a scanning electron micrograph of a red blood cell infected with merozoites. Debris (waste particles) from the preparation of the electron micrograph is also shown.



magnification $\times 23\,000$

Fig. 4.1

- (b) State the difference between the appearance of the red blood cell shown in Fig. 4.1 and a healthy red blood cell.

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- (c) When *Plasmodium* is present within mature red blood cells there is a decrease in concentration of haemoglobin in these infected cells.

Suggest why the concentration of haemoglobin within the infected red blood cells decreases.

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..... [1]

A protein known as merozoite surface protein 1 (MSP1) is found as part of the cell surface membrane of merozoites. All species of *Plasmodium* that cause malaria have MSP1. The protein is used to help *Plasmodium* enter human red blood cells.

(d) MSP1 is composed of a single polypeptide, which is coded for by gene *MSP1*.

Some of the steps occurring in the synthesis of MSP1 by *Plasmodium* are described in Table 4.1. They are **not** listed in the correct sequence.

Table 4.1

step	description
A	tRNA, following amino acid activation, attaches to ribosome
B	mRNA passes through nuclear pores
C	a stop codon is reached and the polypeptide is released
D	DNA double helix unwinds
E	codon–anticodon binding occurs
F	RNA polymerase forms phosphodiester bonds
G	peptide bond formation occurs
H	mRNA attaches to ribosome

Use the steps described in Table 4.1 to complete Table 4.2 to show the correct sequence of events as they would occur in the synthesis of MSP1.

Two of the steps have been added to Table 4.2 for you.

Table 4.2

correct sequence	step
1	
2	
3	
4	H
5	
6	
7	
8	C



- (e) The MSP1 protein has three levels of protein structure: primary, secondary and tertiary.

Outline the changes that occur after translation to a polypeptide, such as the MSP1 polypeptide, that result in a protein showing secondary and tertiary structure.

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- (f) Subunit vaccines are vaccines that contain non-self antigens, but do **not** contain whole organisms. The aim is to stimulate a primary immune response after the vaccine is given so that the person gains artificial active immunity.

One trial that has been carried out on human volunteers has used MSP1 from *Plasmodium falciparum* in a subunit vaccine against malaria.

Explain why a vaccine containing MSP1 provides artificial active immunity to malaria.

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(g) The female *Anopheles* mosquito is the vector of *Plasmodium*.

Discuss the ways in which the vector is controlled to help prevent the transmission of malaria.

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..... [3]

[Total: 18]